Transport phenomena, turbulence and magnetic fields

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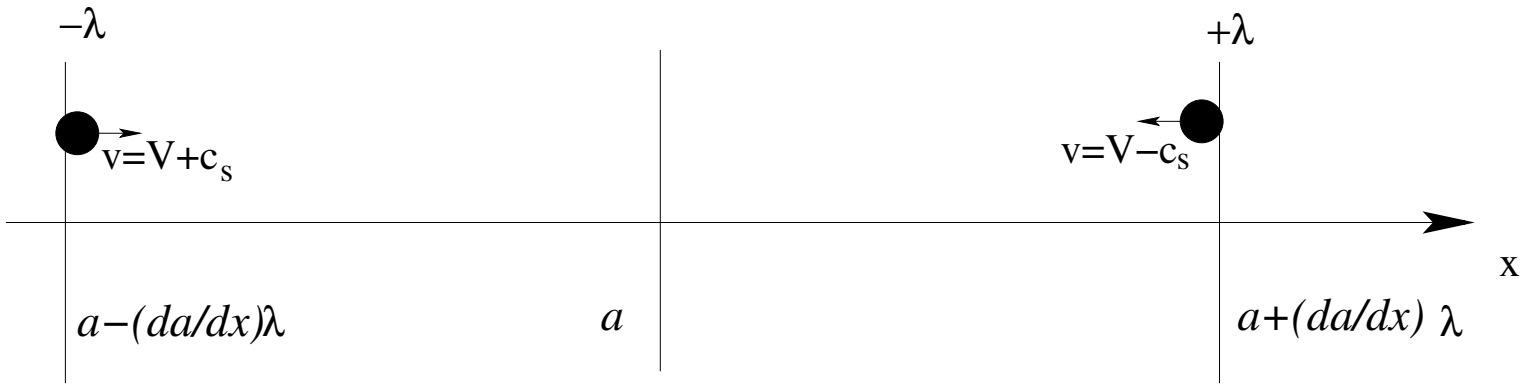
λ is mean free path, c_t is sound velocity (or thermal velocity of electrons, that is about 40 times larger)
Flux and density of quantity a

$$\mathbf{F}_a := \int af(\mathbf{v}, a)\mathbf{v}d\mathbf{v}, \qquad \rho_a := \int af(\mathbf{v}, a)d\mathbf{v}$$



Diffusion-advection (convection-diffusion) equation:

$$\frac{\partial}{\partial t}(\rho_a) + \nabla \cdot (\mathbf{F}_a) = 0$$

$D \sim c_t\lambda$ is diffusion coefficient, $\lambda \sim \frac{1}{\sigma n}$, where σ is interaction cross-section

Fick's law:

$$\mathbf{F}_a = -D\nabla\rho_a + \rho_a\mathbf{V}$$

Heat conduction

Electron thermal conduction:

$$D_{\mathrm{e}} \sim \frac{v_{\mathrm{e}}}{\sigma_{\mathrm{C}} n_{\mathrm{e}}} \sim \frac{(kT)^{5/2}}{e^4 \sqrt{m_{\mathrm{e}}} n_{\mathrm{e}}}$$

Photon diffusion:

$$D_{\gamma} \sim \frac{c}{\sigma_{\gamma} n} \sim \frac{c}{\sigma_{\gamma} n}$$

Degenerate electrons have different $v_{\mathrm{e}} \sim p_{\mathrm{F}}/m_{\mathrm{e}} \gtrsim \sqrt{kT/m_{\mathrm{e}}}$

If $\sigma_{\gamma} = \sigma_T$,

$$\frac{D_{\mathrm{e}}}{D_{\gamma}} \sim 0.1 \left(\frac{T}{10^9 \mathrm{K}} \right)^{5/2}$$

only diffusion coefficients, energy density is divided unequally between radiation and matter

Viscosity

If a is velocity

$$w_{ik} := \int f(\mathbf{v}, a) v_i v_k d\mathbf{v}, \qquad \rho V_i := \int v_i f(\mathbf{v}, a) d\mathbf{v}$$

Momentum (Navier-Stokes) equation

$$\frac{\partial}{\partial t}(V_i) + V_k \nabla_k V_i = -\frac{1}{\rho} \nabla_k w_{ik}$$

Viscous stress:

$$w_{ik} = -\nu\rho \left(\nabla_k V_i + \nabla_i V_k - \frac{2}{3}(\nabla \mathbf{V})\delta_{ik} \right) + \zeta\rho(\nabla \mathbf{V})\delta_{ik}$$

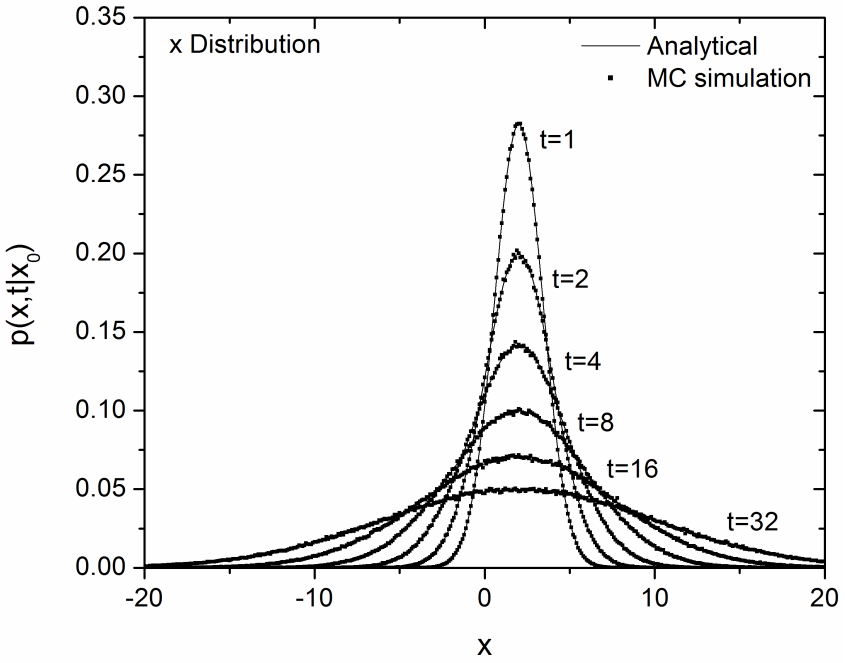
$$w_{ik} = -\nu\rho\nabla_k v_i$$

$\nu \sim \lambda c_{\mathrm{s}}$ is shear kinematic viscosity, ζ is bulk viscosity (not exactly transfer)

Diffusion equation

$$\frac{\partial}{\partial t}(u) + \nabla(D\nabla u) = 0$$

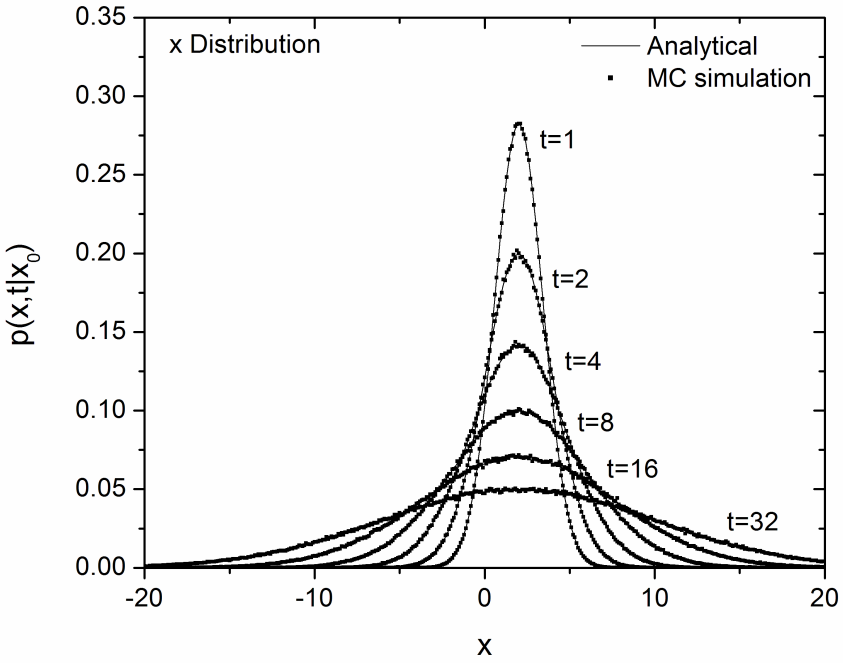
– diffusion equation (violates causality!)



Diffusion equation

$$-\tau \frac{\partial^2}{\partial t^2}(u) + \frac{\partial}{\partial t}(u) + \nabla(D\nabla u) = 0$$

– diffusion with saturation



Microscopic viscosity in accretion discs

Diffusion coefficient for Coulomb-dominated hot ionized discs (ionic viscosity)

$$\nu \sim D \sim c_{\mathrm{t}}\lambda \sim c_{\mathrm{t}}\frac{1}{\sigma n} = \frac{c_{\mathrm{t}}}{\pi n}\left(\frac{kT}{e^2}\right)^2$$

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Reynolds number

$$\frac{\partial}{\partial t}(V_i) + \underbrace{V_k \nabla_k V_i}_{\text{inertial}} = \underbrace{-\frac{1}{\rho}\nabla_k\left(\nu\rho\left(\nabla_k V_i + \nabla_i V_k\right)\right)}_{\text{viscous}}$$

$$\mathrm{Re} \sim \frac{\text{inertial}}{\text{viscous}} \sim= \frac{VL}{\nu} \sim \frac{\text{spatial scale} \times \text{velocity}}{\text{viscosity}}$$

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For an accretion disc, where $L \gtrsim H$ (thickness), $V \lesssim V_{\mathrm{Keplerian}}$,

$$\mathrm{Re} \sim 10^{10} \frac{\Sigma}{10^3 \mathrm{gcm}^{-3}} \frac{V}{0.01c} \left(\frac{T}{10^7 \mathrm{K}}\right)^{-5/2}$$

Turbulence

For $\mathrm{Re} \gtrsim 10^3$ gas is turbulent, and local velocity variations play the role of kinetic motions.
Inertial part of Euler equation:

$$\frac{\partial}{\partial t}(v_i) + v_k \frac{\partial}{\partial x_k}(v_i) = 0$$

Non-linearity transfers energy to smaller scales

$$\sin x \to \sin x \cos x = 0.5 \sin 2x$$

Down to the dissipation (Kolmogorov) scale

$$l_{\mathrm{K}} \sim \nu^{3/4} \varepsilon^{-1/4}$$

Reynolds stress: viscosity without viscosity

Inertial part of Euler equation:

$$\frac{\partial}{\partial t}\left(v_i\right) + v_k \frac{\partial}{\partial x_k}\left(v_i\right) = 0$$

Split velocity in mean and turbulent components $v_i = V_i + \delta v_i$, average $\langle \delta v_i \rangle = 0$

$$\frac{\partial}{\partial t}\left(\langle \delta v_i \delta v_k \rangle\right) = \langle \delta v_i \delta v_j \rangle \frac{\partial}{\partial x_k}\left(V_j\right)$$

More details: Tassoul, “Theory of rotating stars”, or Ogilvie 2003

Viscosity in a shear flow

$$\Omega = \Omega(R)$$

Normal viscosity:

$$w_{r\varphi} = \nu\rho R\frac{d\Omega}{dR}$$

Anisotropic viscosity:

$$w_{r\varphi} = \nu_r\rho\frac{1}{R}\frac{d}{dR}\left(\Omega R^2\right) - 2\nu_\varphi\rho\Omega$$

Conduction

Lorentz force acting on an electron mean free path,

$$v_{\mathrm{e}} \sim \frac{eE}{m_{\mathrm{e}}} \lambda_{\mathrm{e}}$$

Current density

$$\mathbf{j} = \underbrace{\frac{e^2 n_{\mathrm{e}} \lambda_{\mathrm{e}}}{m_{\mathrm{e}}}}_{\text{conductivity}} \mathbf{E}$$

Not dependent on density for a completely ionized plasma; huge (but see later)

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Strong magnetic field

Maxwell equations:

$$(\nabla \cdot \mathbf{B}) = 0$$

$$\nabla \times \mathbf{B} = \frac{1}{c}\frac{\partial}{\partial t}(\mathbf{E}) + \frac{4\pi}{c}\mathbf{j}$$

$$\frac{\partial}{\partial t}(\mathbf{B}) = -c\nabla \times \mathbf{E}$$

($\mathbf{j} = \frac{c}{4\pi}\nabla \times \mathbf{B}$ if displacement current is neglected)

Generalized Ohm's law:

$$\mathbf{j} = \sigma \left(\underbrace{\mathbf{E} + \frac{1}{c}\mathbf{v} \times \mathbf{B}}_{\text{comoving } \mathbf{E}'} + \underbrace{\frac{1}{ecn_{\mathrm{e}}}\mathbf{j} \times \mathbf{B}}_{\text{Hall effect}} \right)$$

- $\mathbf{j} = \sigma \mathbf{E}$ (Ohm's law, $v \ll c$, small σ)
- $\mathbf{E} = -\frac{1}{c}\mathbf{v} \times \mathbf{B}$ (ideal MHD, $\sigma \to \infty$)
- $\mathbf{E} = -\frac{1}{ecn_{\mathrm{e}}}\mathbf{j} \times \mathbf{B}$ (Hall plasma, $\sigma \to \infty$, Hall term dominates)

Dynamic equation:

$$\rho \frac{d\mathbf{v}}{dt} = -\nabla p + \frac{1}{c}\mathbf{j} \times \mathbf{B}$$

Maxwell stress tensor

$$w_{ik}^{\text{magnetic}} = \frac{1}{4\pi}B_i B_k - \frac{1}{8\pi}B^2 \delta_{ik}$$

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The role of conductivity

Assuming $\frac{\partial}{\partial t}(\mathbf{E}) = 0$, and $\mathbf{j} = \sigma \mathbf{B}$ leads to diffusion-advection equation for $\mathbf{B}$

$$\frac{\partial}{\partial t}(\mathbf{B}) = \nabla \times [\mathbf{v} \times \mathbf{B}] + \frac{c^2}{4\pi\sigma} \nabla^2 \mathbf{B}$$

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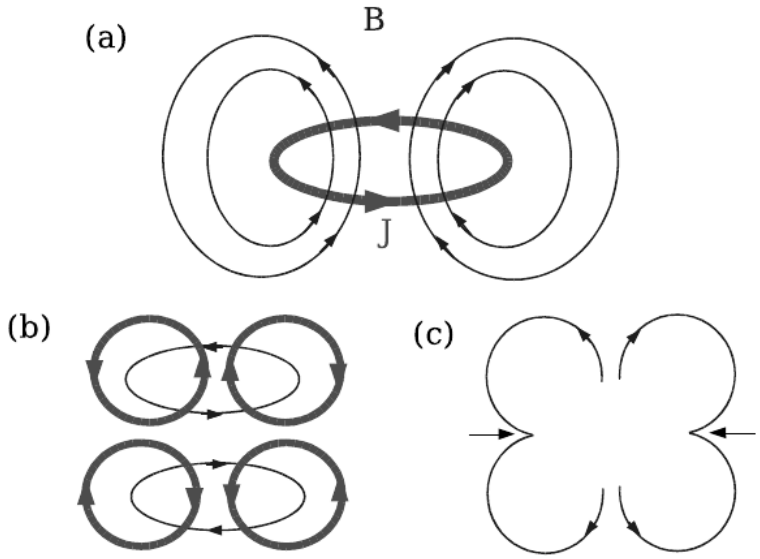
Magnetic Reynolds number:

$$\mathrm{Re_M} = \frac{4\pi\sigma}{c^2} VL$$

The fate of strong magnetic fields



Magnetic field evolution equation

$$\frac{\partial}{\partial t}(\mathbf{B}) \simeq \frac{c}{4\pi e n_{\mathrm{e}}} \nabla \times [\mathbf{B} \times [\nabla \times \mathbf{B}]]$$

Non-linear $\Rightarrow$ produces higher-wavenumber harmonics $\Rightarrow$ increasing $\nabla^2 \mathbf{B}$ until dissipation starts.
Hall evolution time

$$t_{\mathrm{Hall}} \sim \frac{4\pi e n_{\mathrm{e}} L^2}{cB}$$

How does turbulent transport work in astrophysics: